

Research Notes on Semi-supervised learning

Semi-supervised learning

>> Section 1

Generative models Generative approaches to statistical learning first seek to estimate $p(x|y)$, the distribution of data points belonging to each class. The probability $p(y|x)$ that a given point x has label y is then proportional to $p(x|y)p(y)$ by Bayes' rule. Semi-supervised learning with generative models can be viewed either as an extension of supervised learning (classification plus information about $p(x)$) or as an extension of unsupervised learning (clustering plus some labels). Generative models assume that the distributions take some particular form $p(x|y, \theta)$ parameterized by the vector θ . If these assumptions are incorrect, the unlabeled data may actually decrease the accuracy of the solution relative to what would have been obtained from labeled data alone. However, if the assumptions are correct, then the unlabeled data necessarily improves performance. The unlabeled data are distributed according to a mixture of individual-class distributions. In order to learn the mixture distribution from the unlabeled data, it must be identifiable, that is, different parameters must yield different summed distributions. Gaussian mixture distributions are identifiable and commonly used for generative models. The parameterized joint distribution can be written as $p(x, y | \theta) = p(y | \theta) p(x | y, \theta)$ by using the chain rule. Each parameter vector θ is associated with a decision function $f_{\theta}(x) = \underset{y}{\operatorname{argmax}} p(y | x, \theta)$. The parameter is then chosen based on fit to both the labeled and unlabeled data, weighted by λ :

>> Section 2

The data lie approximately on a manifold of much lower dimension than the input space. In this case learning the manifold using both the labeled and unlabeled data can avoid the curse of dimensionality. Then learning can proceed using distances and densities defined on the manifold. The manifold assumption is practical when high-dimensional data are generated by some process that may be hard to model directly, but which has only a few degrees of freedom. For instance, human voice is controlled by a few vocal folds, and images of various facial expressions are controlled by a few muscles. In these cases, it is better to consider distances and smoothness in the natural space of the generating problem, rather than in the space of all possible acoustic waves or images, respectively.

>> Section 3

An exact solution is intractable due to the non-convex term $(1 - |f(x)|) +$, so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).

>> Section 4

where H is a reproducing kernel Hilbert space and M is the manifold on which the data lie. The regularization parameters λ_A and λ_I control smoothness in the ambient and intrinsic spaces respectively. The graph is used to approximate the intrinsic regularization term. Defining the graph Laplacian $L = D - W$ where $D_{ii} = \sum_j W_{ij}$ and f is the vector $[f(x_1) \dots f(x_{l+u})]$, we have

>> Section 5

$$\operatorname{argmin}_{f \in H} \left(\frac{1}{2} \sum_{i=1}^l V(f(x_i), y_i) + \lambda_A \int_M |\nabla f(x)|^2 dp(x) + \lambda_I \int_M |\nabla f(x)|^2 dp(x) \right)$$

>> Section 6

In human cognition Human responses to formal semi-supervised learning problems have yielded varying conclusions about the degree of influence of the unlabeled data. More natural learning problems may also be viewed as instances of semi-supervised learning. Much of human concept learning involves a small amount of direct instruction (e.g. parental labeling of objects during childhood) combined with large amounts of unlabeled experience (e.g. observation of objects without naming or counting them, or at least without feedback). Human infants are sensitive to the structure of unlabeled natural categories such as images of dogs and cats or male and female faces. Infants and children take into account not only unlabeled examples, but the sampling process from which labeled examples arise.

>> Section 7

$$f^T L f = \sum_{i,j=1}^{l+u} W_{ij} (f_i - f_j)^2 \approx \int_M |\nabla f(x)|^2 dp(x) = \sum_{i,j=1}^{l+u} W_{ij} (f_i - f_j)^2 \approx \int_M |\nabla f(x)|^2 dp(x)$$
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graph-based approach to Laplacian regularization is to put in relation with finite difference method. The Laplacian can also be used to extend the supervised learning algorithms: regularized least squares and support vector machines (SVM) to semi-supervised versions Laplacian regularized least squares and Laplacian SVM.

>> Section 8

The acquisition of labeled data for a learning problem often requires a skilled human agent (e.g. to transcribe an audio segment) or a physical experiment (e.g. determining the 3D structure of a protein or determining whether there is oil at a particular location). The cost associated with the labeling process thus may render large, fully labeled training sets infeasible, whereas acquisition of unlabeled data is relatively inexpensive. In such situations, semi-supervised learning can be of great practical value. Semi-supervised learning is also of theoretical interest in machine learning and as a model for human learning.